

# Driver Drowsiness Detection Using Deep Learning-Based Eye State Classification - Report

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**Abstract**—Driver drowsiness is one of the major causes of road accidents worldwide. Detecting drowsiness in real-time can significantly reduce the number of accidents and improve road safety. This project presents a deep learning-based driver drowsiness detection system using Convolutional Neural Networks (CNNs) and real-time eye state identification. The proposed system focuses on the eye region because eye fatigue and prolonged eye closure are among the earliest signs of drowsiness. MediaPipe Face Mesh is used for facial landmark detection and Region of Interest (ROI) extraction. Three transfer learning architectures, namely InceptionV3, VGG16, and ResNet50V2, are used for classification of eye states into drowsy and non-drowsy categories. Experimental analysis shows that ResNet50V2 achieves the highest accuracy among the three models. The proposed approach demonstrates high reliability and effectiveness for real-time driver monitoring systems.

**Index Terms**—Driver Drowsiness Detection, Deep Learning, Convolutional Neural Network, MediaPipe, Transfer Learning, ResNet50V2, Computer Vision.

## I. INTRODUCTION

Road accidents caused by driver fatigue and drowsiness have become a serious issue worldwide. Long driving hours, lack of sleep, stress, and night driving are common reasons that reduce the alertness of drivers. According to road safety reports, a significant percentage of traffic accidents occur because drivers lose concentration or fall asleep while driving.

Traditional methods for detecting drowsiness include steering behavior analysis, vehicle lane tracking, and physiological signal monitoring such as Electroencephalogram (EEG) and Electrocardiogram (ECG). Although these methods provide accurate results, many of them are invasive and uncomfortable for drivers.

With the advancement of Artificial Intelligence and Computer Vision, image-based drowsiness detection systems have become highly popular. Facial features such as eye blinking, eye closure duration, yawning, and head movements provide important information regarding the driver's alertness level. Among these features, eye state analysis is one of the most reliable indicators of drowsiness.

This project presents a CNN-based driver drowsiness detection system using real-time eye state identification. The proposed system uses MediaPipe Face Mesh to detect facial landmarks and extract the eye region. Transfer learning models including InceptionV3, VGG16, and ResNet50V2 are trained on the extracted eye images to classify the driver as drowsy or non-drowsy.

The objectives of this project are as follows:

- To detect driver drowsiness using deep learning techniques.
- To extract eye regions accurately using facial landmark detection.
- To compare the performance of different CNN architectures.
- To implement a real-time drowsiness detection mechanism.

## II. LITERATURE REVIEW

Several researchers have proposed methods for detecting driver drowsiness using machine learning and deep learning techniques.

Park et al. proposed a deep drowsiness detection architecture using RGB facial videos. Their system used VGG-FaceNet and achieved moderate performance in real-time scenarios.

Chirra et al. used Haar Cascade classifiers for eye detection and CNNs for classification. Their approach achieved high validation accuracy but suffered from limitations during head movements.

Zhao et al. proposed a system based on facial landmark extraction and CNN classification. They used MTCNN for feature extraction and obtained high classification performance.

Phan et al. used Dlib facial landmarks combined with MobileNetV2 and ResNet50V2 for drowsiness detection. Their work demonstrated the effectiveness of transfer learning models.

Recent research has shown that CNN architectures such as ResNet50V2, VGG16, MobileNet, and InceptionV3 are highly effective for image classification tasks. These architectures are capable of extracting deep spatial features from facial images and provide high classification accuracy.

The literature review indicates that focusing on the eye region provides better drowsiness prediction compared to full-face analysis. Therefore, this project mainly focuses on eye state detection.

## III. PROBLEM STATEMENT

Driver drowsiness is a major cause of road accidents and fatalities. Existing traditional systems are either expensive, invasive, or inaccurate under varying environmental conditions. Therefore, there is a need for a non-invasive, real-time, and highly accurate drowsiness detection system.

The main problem addressed in this project is the development of a robust deep learning-based system capable of identifying drowsiness from eye states in real time.

#### IV. PROPOSED METHODOLOGY

The proposed methodology consists of multiple stages including video acquisition, preprocessing, facial landmark detection, ROI extraction, CNN training, and real-time prediction.

##### A. System Architecture

The overall workflow of the proposed system is shown below:

- 1) Video acquisition from webcam or dataset.
- 2) Frame extraction from videos.
- 3) Facial landmark detection using MediaPipe.
- 4) Eye Region of Interest (ROI) extraction.
- 5) Image preprocessing and normalization.
- 6) CNN-based eye state classification.
- 7) Drowsiness detection based on eye closure duration.

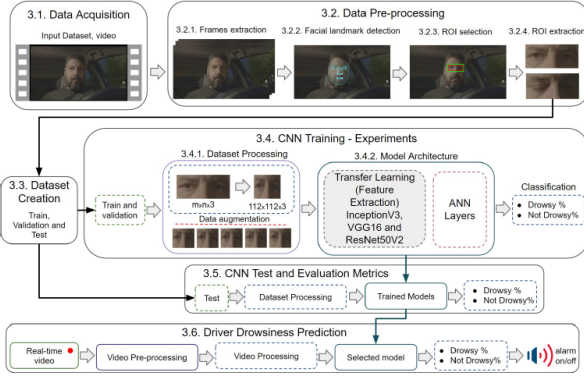


Fig. 1. Overall System Architecture

##### B. Data Acquisition

The dataset used in this project is the NITYMED dataset, which contains videos of drivers under different drowsiness conditions. The dataset includes both male and female subjects driving in real-world night-time environments.

The videos contain different facial expressions, eye movements, and head poses, making the dataset suitable for realistic drowsiness detection.

##### C. Frame Extraction

Frames are extracted from videos at a fixed frame rate. Each extracted frame is processed individually for facial landmark detection.

If the video frame rate is represented as:

$$F = fps \times duration \quad (1)$$

where:

- $fps$  = frames per second
- $duration$  = duration of video in seconds

For example, a 30-second video with 25 fps generates:

$$F = 25 \times 30 = 750 \text{ frames} \quad (2)$$

##### D. Facial Landmark Detection

MediaPipe Face Mesh is used to detect 468 facial landmarks in real time. The landmarks help identify important facial regions such as eyes, nose, and mouth.

MediaPipe provides high accuracy and robustness even when the driver changes head orientation.

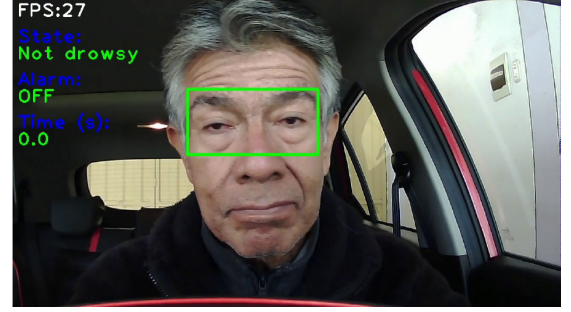


Fig. 2. Facial Landmark Detection using MediaPipe

##### E. ROI Extraction

The eye region is extracted using selected landmark points around the eyes. ROI correction is performed to ensure proper extraction during head movements.

The extracted ROI contains both eyes and is resized before being passed to CNN models.

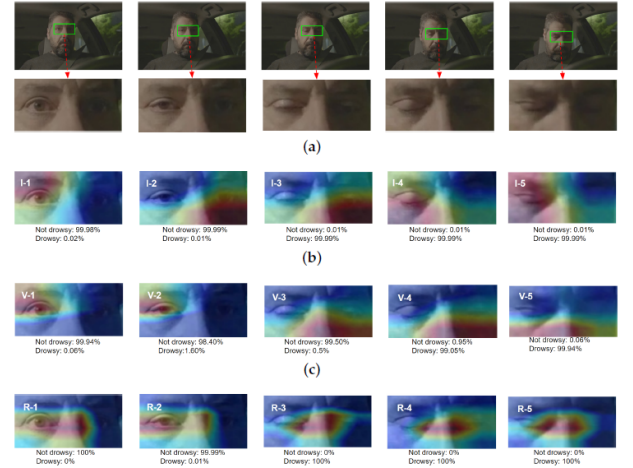


Fig. 3. Eye Region Extraction

##### F. Dataset Preparation

The extracted eye images are categorized into two classes:

- Drowsy
- Non-Drowsy

The dataset is divided into training, validation, and testing sets.

TABLE I  
DATASET DISTRIBUTION

| Dataset    | Drowsy | Non-Drowsy |
|------------|--------|------------|
| Training   | 2380   | 2380       |
| Validation | 510    | 510        |
| Testing    | 510    | 510        |

### G. Image Preprocessing

The extracted eye images are resized to  $112 \times 112 \times 3$  pixels. Normalization is performed by dividing pixel values by 255:

$$X_{norm} = \frac{X}{255} \quad (3)$$

Data augmentation techniques such as rotation, flipping, and zooming are applied to avoid overfitting.

## V. CNN ARCHITECTURES

Three transfer learning architectures are used in this project:

### A. InceptionV3

InceptionV3 is a deep CNN architecture developed by Google. It uses inception modules to reduce computational complexity while maintaining high accuracy.

Advantages of InceptionV3:

- Efficient computation
- Reduced parameters
- High classification accuracy

### B. VGG16

VGG16 is a widely used CNN architecture consisting of 16 deep layers. It uses small convolution filters and deep feature extraction.

Advantages of VGG16:

- Simple architecture
- Strong feature extraction
- Effective for image classification

### C. ResNet50V2

ResNet50V2 uses residual connections to solve the vanishing gradient problem. It enables training of very deep neural networks.

Advantages of ResNet50V2:

- High accuracy
- Residual learning
- Better gradient flow

## VI. MODEL TRAINING

Transfer learning is used to train the CNN architectures. Pretrained ImageNet weights are utilized for feature extraction.

The fully connected layer is modified for binary classification.

Training parameters used are shown in Table II.

TABLE II  
TRAINING HYPERPARAMETERS

| Parameter           | Value |
|---------------------|-------|
| Optimizer           | Adam  |
| Learning Rate       | 0.001 |
| Epochs              | 30    |
| Batch Size          | 32    |
| Dropout             | 0.3   |
| Activation Function | ReLU  |

### A. Loss Function

Categorical Cross Entropy loss is used for binary classification.

$$Loss = - \sum y \log(\hat{y}) \quad (4)$$

### B. Evaluation Metrics

The following metrics are used for evaluation:

- Accuracy
- Precision
- Recall
- F1-Score

Accuracy is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

Precision is calculated as:

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

Recall is calculated as:

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

F1-Score is calculated as:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (8)$$

## VII. EXPERIMENTAL RESULTS

The models are evaluated using testing data. Among all architectures, ResNet50V2 achieved the best performance.

TABLE III  
PERFORMANCE COMPARISON OF CNN MODELS

| Model       | Accuracy | Recall | F1-Score |
|-------------|----------|--------|----------|
| InceptionV3 | 99.31%   | 99.08% | 99.27%   |
| VGG16       | 99.41%   | 99.37% | 99.39%   |
| ResNet50V2  | 99.71%   | 99.47% | 99.71%   |

### A. Confusion Matrix

The confusion matrix shows the classification performance of the CNN models.

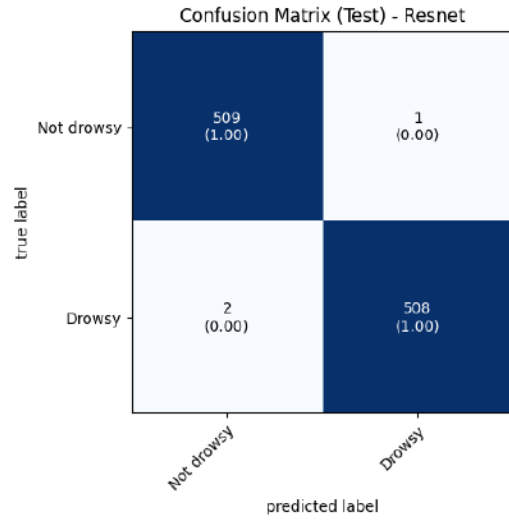


Fig. 4. Confusion Matrix

### B. Accuracy and Loss Graphs

Training and validation accuracy/loss graphs are plotted during model training.

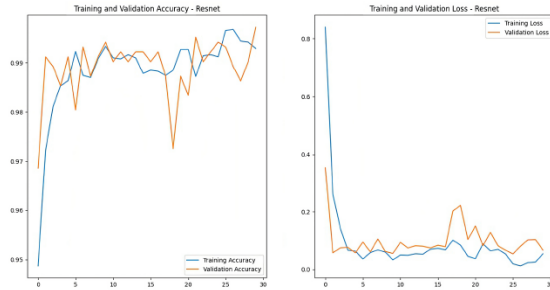


Fig. 5. Training Accuracy and Loss

## VIII. REAL-TIME DROWSINESS DETECTION

The trained model is integrated into a real-time monitoring system.

The webcam captures video frames continuously. MediaPipe extracts the eye region and the CNN predicts whether the eyes are open or closed.

If the eye closure duration exceeds 300 milliseconds, the system triggers an alarm.

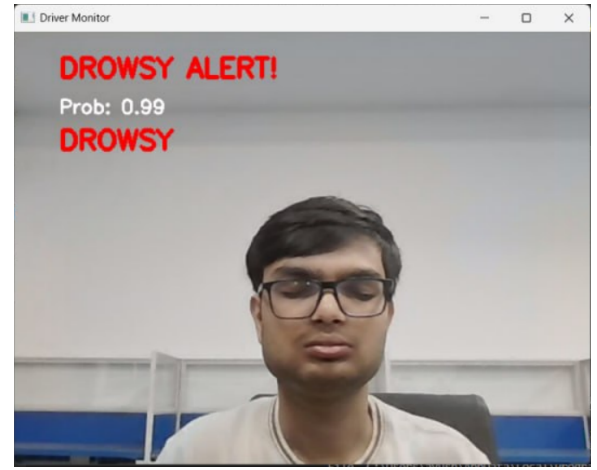


Fig. 6. Real-Time Driver Drowsiness Detection

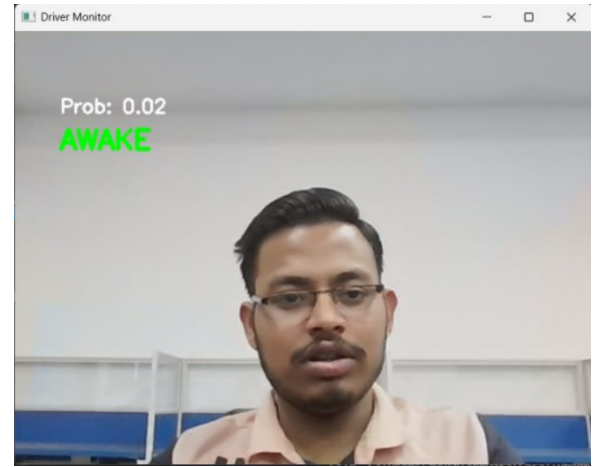


Fig. 7. Real-Time Driver Drowsiness Detection

## IX. ADVANTAGES OF PROPOSED SYSTEM

The proposed system provides several advantages:

- Non-invasive drowsiness detection
- High classification accuracy
- Real-time monitoring capability
- Robust eye detection using MediaPipe
- Effective under varying head poses

## X. APPLICATIONS

The proposed system can be used in:

- Smart vehicles
- Driver assistance systems
- Transportation safety systems
- Fleet monitoring systems
- Autonomous vehicle safety modules

## XI. LIMITATIONS

Although the proposed system provides excellent performance, it still has certain limitations:

- Performance may reduce under poor lighting conditions.

- Glasses and occlusions can affect detection accuracy.
- High computational requirements for deep CNN models.

## XII. FUTURE SCOPE

Future improvements can include:

- Integration of infrared cameras for night vision.
- Addition of yawning detection.
- Deployment on embedded systems.
- Use of lightweight CNN models for mobile devices.
- Multimodal fatigue detection using EEG and facial analysis.

## XIII. CONCLUSION

This project presents a deep learning-based driver drowsiness detection system using CNN architectures and real-time eye state identification. MediaPipe Face Mesh is used for accurate facial landmark detection and eye ROI extraction.

Three CNN architectures, namely InceptionV3, VGG16, and ResNet50V2, are evaluated using transfer learning. Experimental results demonstrate that ResNet50V2 achieves the highest classification accuracy and robustness.

The proposed approach provides a reliable and non-invasive solution for real-time driver monitoring systems and can help reduce accidents caused by driver fatigue.

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### Code Repository:

<https://github.com/Chitranshu-DTU-Official/Deep-Learning-Project>